HDOC Day-13 Activity

Question

Write a **menu-driven C program using switch-case** to input a positive integer and count the number of **even digits** and **odd digits**.

Example:
Number = 12045
Even_dig = 3
Odd_dig = 2

Note: 0 is an even digit.

1. Algorithm

1. Start.
2. Declare variables num, temp, digit, even = 0, odd = 0, and choice.
3. Display the menu.
4. Enter the user's choice.
5. Use switch(choice).
6. For case 1:
   - Input a positive integer.
   - Store the number in temp.
   - Repeat while temp > 0:
     - Find the last digit using digit = temp % 10.
     - If digit % 2 == 0, increase even by 1.
     - Otherwise, increase odd by 1.
     - Remove the last digit using temp = temp / 10.
   - Display the number of even and odd digits.
7. For any other choice, display "Invalid choice".
8. Stop.

2. Test Case Table

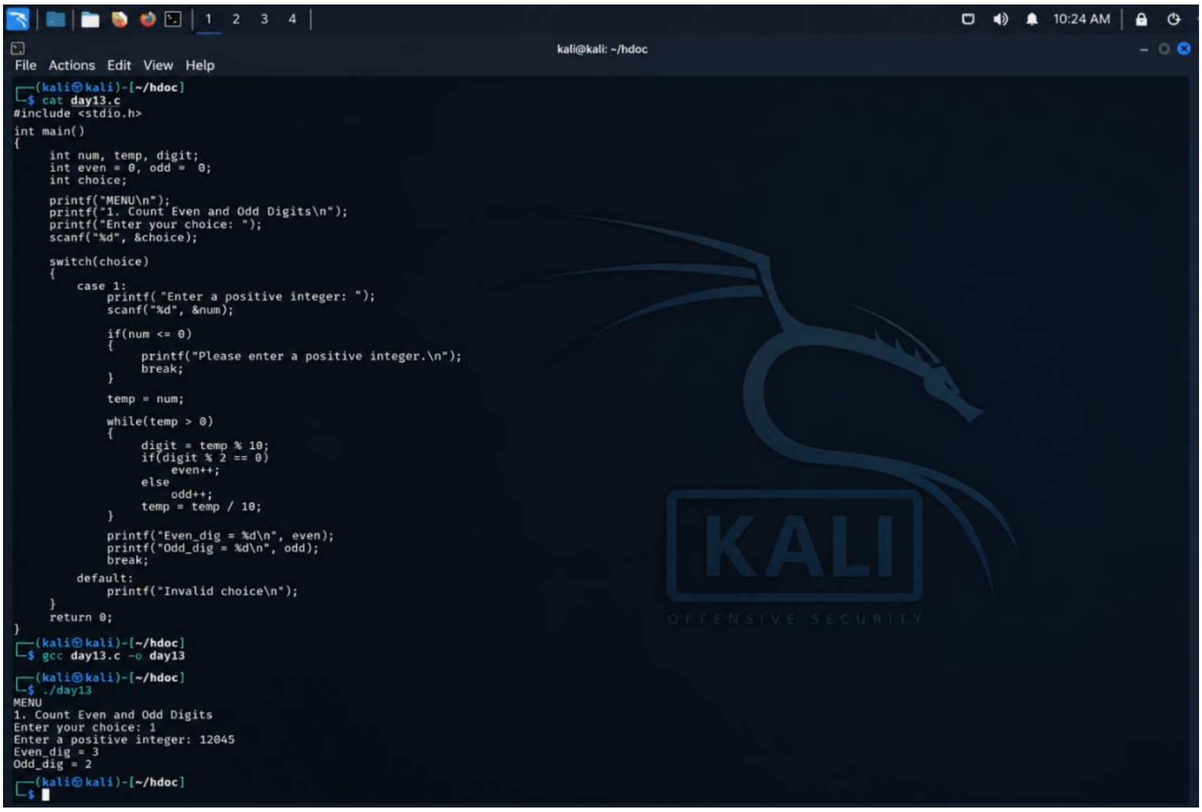
Test Case	Input Number	Expected Output	Actual Output	Result
1	12045	Even_dig = 3, Odd_dig = 2	Even_dig = 3, Odd_dig = 2	Pass
2	2468	Even_dig = 4, Odd_dig = 0	Even_dig = 4, Odd_dig = 0	Pass
3	13579	Even_dig = 0, Odd_dig = 5	Even_dig = 0, Odd_dig = 5	Pass
4	123456	Even_dig = 3, Odd_dig = 3	Even_dig = 3, Odd_dig = 3	Pass
5	90817	Even_dig = 2, Odd_dig = 3	Even_dig = 2, Odd_dig = 3	Pass

3. C Program

```
#include <stdio.h>

int main()
{
  int num, temp, digit;
  int even = 0, odd = 0;
  int choice;

  printf("MENU\n");
  printf("1. Count Even and Odd Digits\n");
  printf("Enter your choice: ");
  scanf("%d", &choice);

  switch(choice)
  {
    case 1:
      printf("Enter a positive integer: ");
      scanf("%d", &num);

      if(num <= 0)
```

```
        {
            printf("Please enter a positive integer.\n");
            break;
        }

        temp = num;

        while(temp > 0)
        {
            digit = temp % 10;

            if(digit % 2 == 0)
                even++;
            else
                odd++;

            temp = temp / 10;
        }

        printf("Even_dig = %d\n", even);
        printf("Odd_dig = %d\n", odd);
        break;

    default:
        printf("Invalid choice\n");
}

return 0;
```

}



```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/hdoc]
└─$ cat day13.c
#include <stdio.h>
int main()
{
     int num, temp, digit;
     int even = 0, odd =  0;
     int choice;
     printf("MENU\n");
     printf("1. Count Even and Odd Digits\n");
     printf("Enter your choice: ");
     scanf("%d", &choice);
     switch(choice)
     {
         case 1:
             printf( "Enter a positive integer: ");
             scanf("%d", &num);
             if(num <= 0)
             {
                  printf("Please enter a positive integer.\n");
                  break;
             }
             temp = num;
             while(temp > 0)
             {
                  digit = temp % 10;
                  if(digit % 2 == 0)
                      even++;
                  else
                      odd++;
                  temp = temp / 10;
             }
             printf("Even_dig = %d\n", even);
             printf("Odd_dig = %d\n", odd);
             break;
         default:
             printf("Invalid choice\n");
     }
     return 0;
}
┌──(kali㉿kali)-[~/hdoc]
└─$ gcc day13.c -o day13
┌──(kali㉿kali)-[~/hdoc]
└─$ ./day13
MENU
1. Count Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 12045
Even_dig = 3
Odd_dig = 2
┌──(kali㉿kali)-[~/hdoc]
└─$
```

For 12045, the even digits are **0, 2, 4** → 3, and the odd digits are **1, 5** → 2.